



AZ-700

Design and Implement Hybrid Networking

AZ-700 Agenda

Module 01: Introduction to Azure Virtual Networks

Module 02: Designing and Implementing Hybrid Networking ←

Module 03: Designing and Implementing Azure ExpressRoute

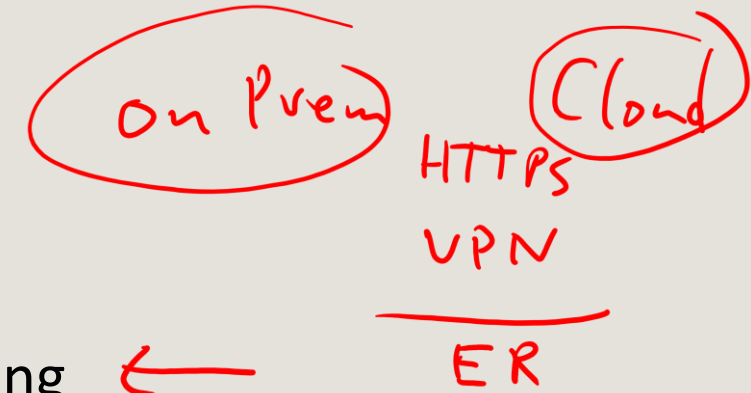
Module 04: Load balance non-HTTP(S) traffic in Azure

Module 05: Load balance HTTP(S) traffic in Azure

Module 06: Design and Implement Network Security

Module 07: Design and Implement private access to Azure Services

Module 08: Design and Implement Network Monitoring



Module Overview

- Design and implement Azure VPN Gateway 42 min
- Exercise – Create and configure a Virtual Network Gateway Lab ✓
- Connect networks with Site-to-site VPN connections
- Connect devices to networks with Point-to-site VPN connections
- Connect remote resources by using Azure Virtual WANs
- Exercise – Create a Virtual WAN by using the Azure Portal
- Create a network virtual appliance (NVA) in a virtual hub

Design and implement Azure VPN gateway

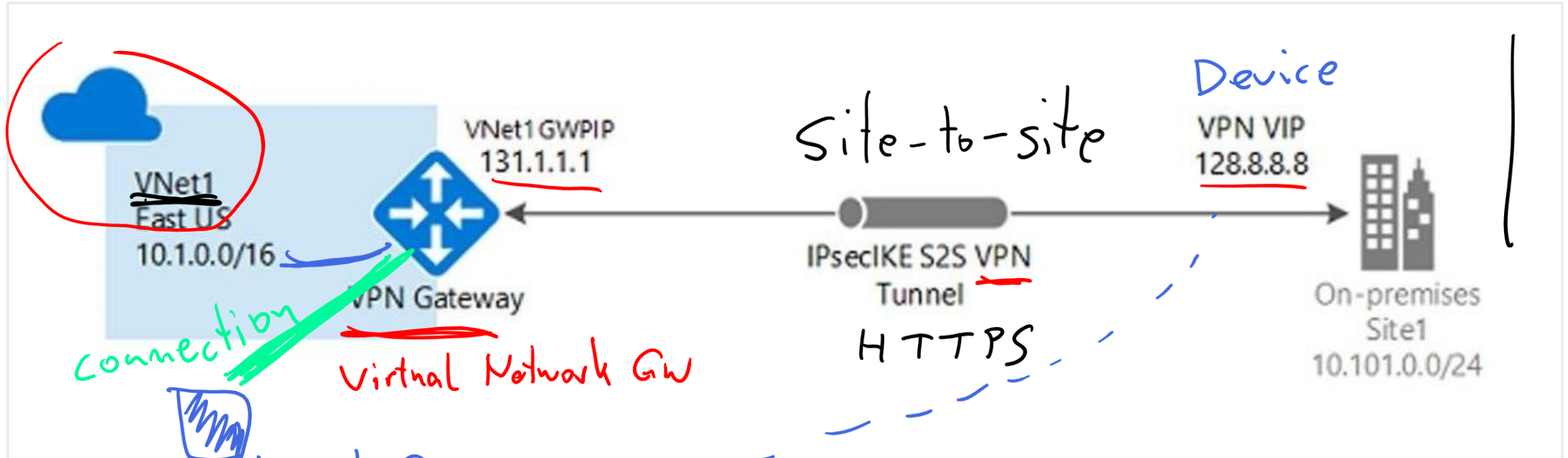


Learning Objectives – Azure VPN Gateway

- Plan a VPN Gateway
- Create the Gateway Subnet
- VPN Gateway Configuration requirements
- VPN Gateway Types
- Choose the appropriate Gateway SKU and Generation
- Create the Local Network Gateway
- Configure the on-premises VPN device
- Create the VPN connection
- Verify and troubleshoot the VPN connection
- High availability options for VPN connections
- Demonstration
- Learning Recap

Plan a VPN Gateway

SOTA



Site-to-site connections connect on-premises datacenters to Azure virtual networks

VNet-to-VNet connections connect Azure virtual networks to each other

Point-to-site (User VPN) connections connect individual devices to Azure virtual networks

Create the Gateway Subnet

The gateway subnet contains the IP addresses; if possible, Use a CIDR block of /27 or larger

When you create your gateway subnet, gateway VMs are deployed to the gateway subnet and configured with the required VPN gateway settings

Never deploy other resources (for example, additional VMs) to the gateway subnet

The screenshot shows the Azure portal interface for creating a new subnet. The top bar indicates the virtual network is 'vnet01 - Subnets'. Below the search bar, there are buttons for '+ Subnet' and '+ Gateway subnet', with the latter being highlighted by a red rectangle. A 'Refresh' button is also present. The 'Add subnet' dialog is open, showing the following configuration:

- Name:** GatewaySubnet
- Subnet address range:** 10.0.32/28 (10.0.32 - 10.0.47 (11 + 5 Azure reserved addresses))
- Add IPv6 address space:** ☐
- NAT gateway:** None
- Network security group:** None
- Route table:** None
- SERVICE ENDPOINTS:** Create service endpoint policies to allow traffic to specific azure resources from your virtual network over service endpoints. [Learn more](#). Services: 0 selected.
- SUBNET DELEGATION:** Delegate subnet to a service: None.
- NETWORK POLICY FOR PRIVATE ENDPOINTS:** The network policy affects all private endpoints in this subnet. Select the types of network policies that control traffic going to the private endpoints in this subnet. [Learn more](#).

At the bottom of the dialog are 'Save' and 'Cancel' buttons.

VPN Gateway Configuration requirements

Most VPN types are Route-based

Your choice of gateway SKU affects the number of connections you can have and the aggregate throughput benchmark

Associate a virtual network that includes the gateway subnet

The gateway needs a public IP address

[Home](#) > [Virtual network gateways](#) >

Create virtual network gateway ...

Instance details

Gateway type * ⓘ

☒ VPN ☐ ExpressRoute

SKU * ⓘ

VpnGw2

Generation ⓘ

Generation2

Virtual network * ⓘ

[Create virtual network](#)

Public IP address

Public IP address * ⓘ

☒ Create new ☐ Use existing

Public IP address name *

Public IP address SKU

Standard

Assignment

☐ Dynamic ☒ Static

Enable active-active mode *

☒ Enabled ☐ Disabled



It can take up to 45 minutes to provision the VPN gateway

Choose the appropriate Gateway SKU and Generation

Sampling of available SKUs

SKU * ⓘ

Generation ⓘ

Gen	SKU	S2S/VNet-to-VNet Tunnels	P2S IKEv2 Connections	Throughput Benchmark
1	VpnGw1Az	Max. 30	Max. 250	650 Mbps
1	VpnGw2Az	Max. 30	Max. 500	1.0 Gbps
2	VpnGw2Az	Max. 30	Max. 500	1.25 Gbps
1	VpnGw3Az	Max. 30	Max. 1000	1.25 Gbps
2	VpnGw3Az	Max. 30	Max. 1000	2.5 Gbps
2	VpnGw4Az	Max. 100	Max. 5000	5.0 Gbps

The Gateway SKU affects the connections and the throughput

Resizing is allowed within the generation

The Basic SKU (not shown) is legacy and should not be used

Create the Local Network Gateway

Reflects the on-premises network configuration and enables Azure to route to your on-premises network

Give the site a name by which Azure can refer to it

Use a public IP address or FQDN for Local Network Gateway Endpoint

Specify the IP address prefixes that will be routed through the gateway to the VPN device

Create local network gateway

Name *

VNet1LocalNet



Endpoint ⓘ

IP address

FQDN

IP address * ⓘ

33.2.1.5



Address space ⓘ

192.168.3.0/24



Add additional address range



☐ Configure BGP settings

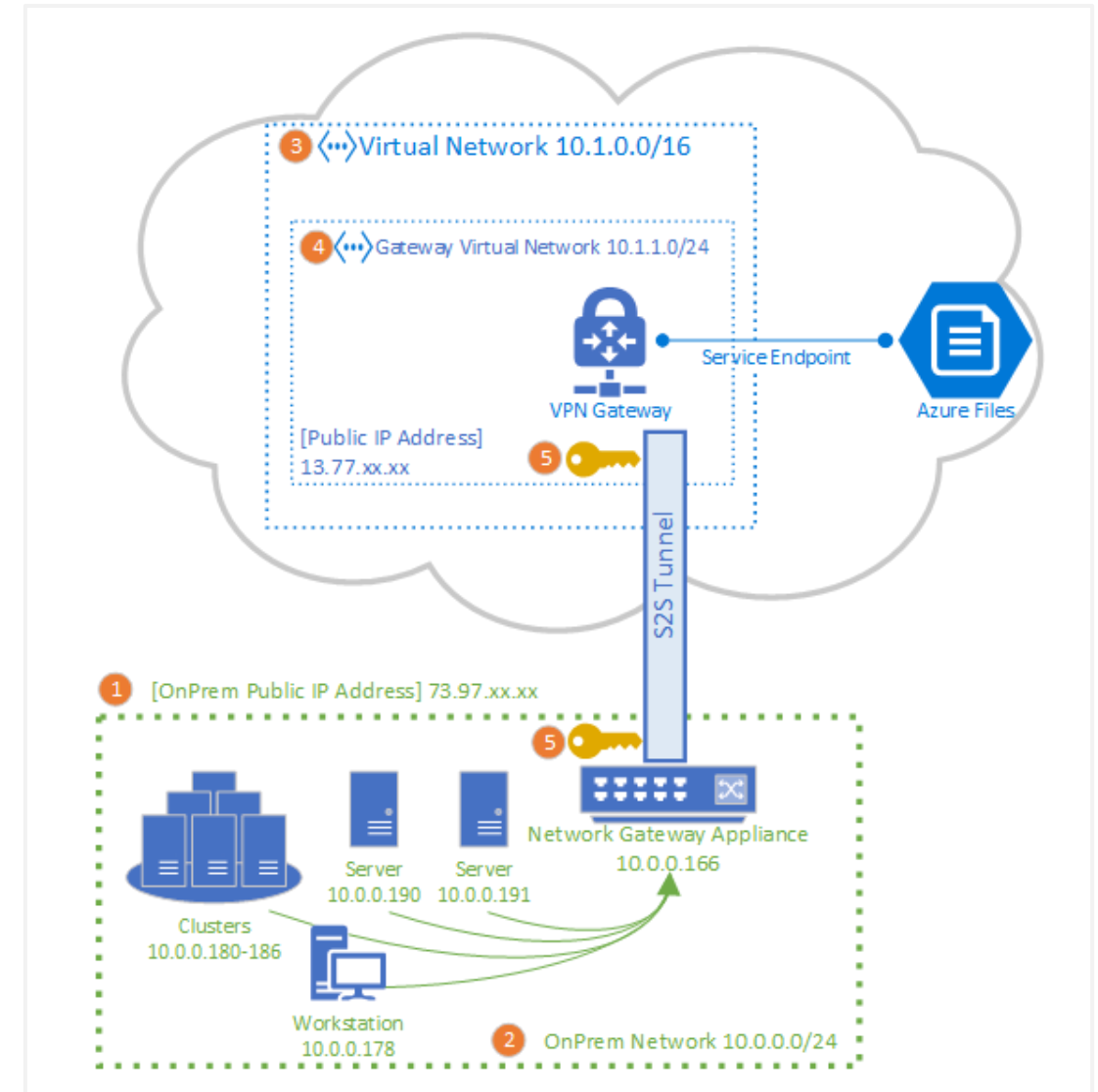
Configure the On-premises VPN Device

Remember the shared key for the Azure connection (next step)

Consult the list of supported VPN devices (Cisco, Juniper, Ubiquiti, Barracuda Networks)

Specify the public IP address (previous step)

A VPN device configuration script may be available



Create the VPN Connection

Once your VPN gateway is created and the on-premises device is configured, create a connection object

Configure a name for the connection and specify the type as Site-to-site (IPsec)

Select the VPN gateway and the Local Network Gateway

Enter the Pre-Shared key for the connection

 **Add connection** ×

vng01

Name *

Azure-to-OnPrem ✓

Connection type ⓘ

Site-to-site (IPsec) ∨

***Virtual network gateway** ⓘ 🔒

vng01

***Local network gateway** ⓘ >

Azure-to-OnPrem

Shared key (PSK) * ⓘ

abc123 ✓

Choose local network gat... □ ×

+ Create new

 Azure-to-OnPrem NetworkRG

Verify and troubleshoot the VPN connection

Validate VPN throughput to a VNet

Utilize Network Watcher

Troubleshoot Azure VPN Gateway using diagnostic logs

Check UDR and NSGs on the gateway subnet

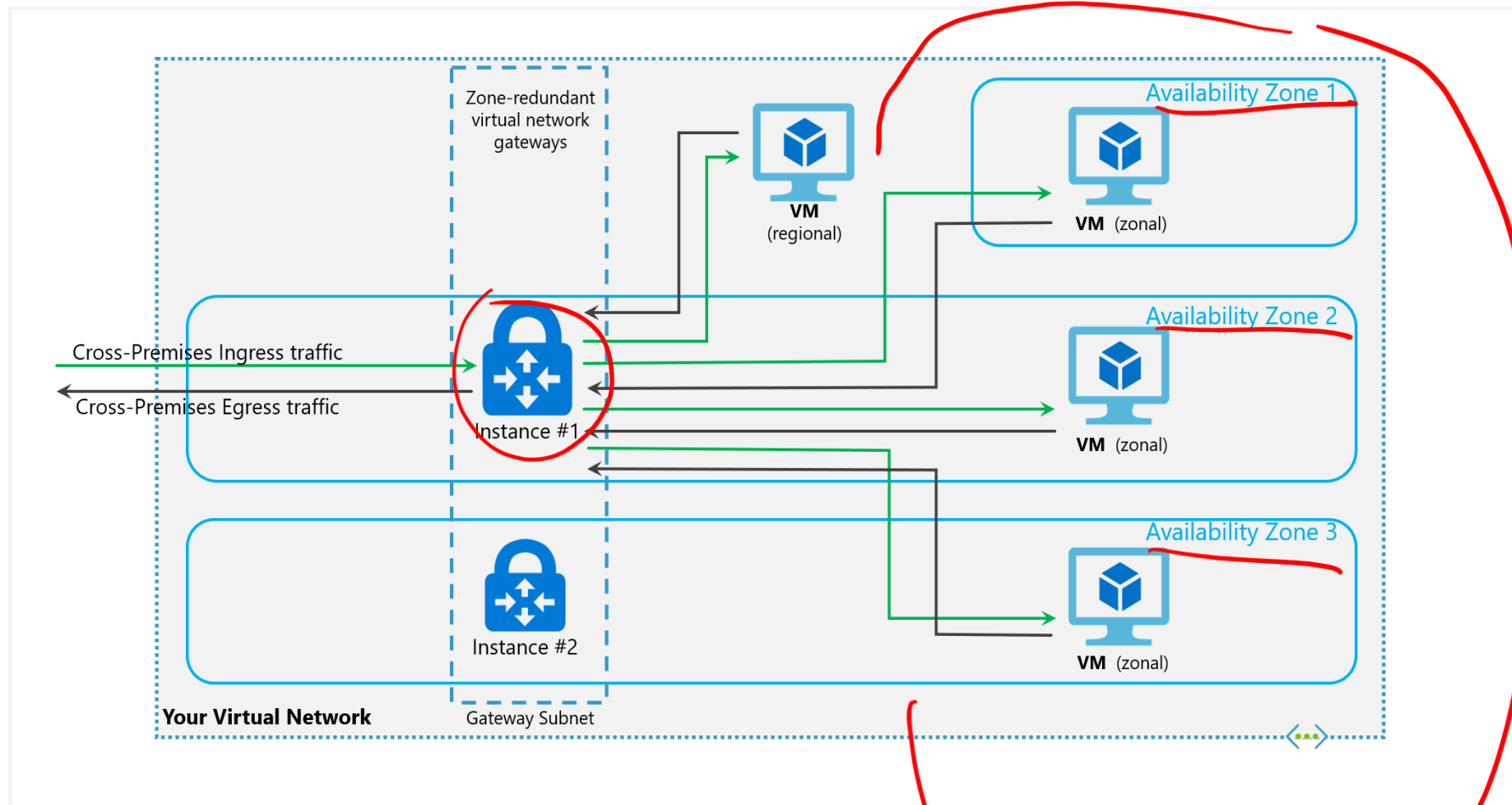
Check whether the on-premises VPN device is validated

Verify the Azure gateway health probe

Verify the shared key and the VPN peer IPs

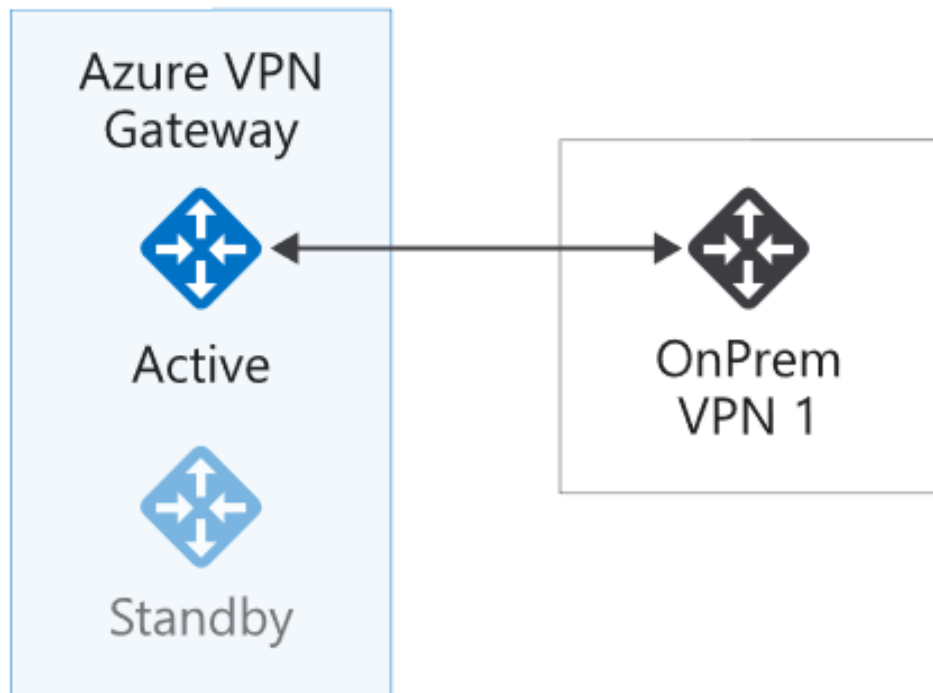
Check whether the on-premises VPN device has the perfect forward secrecy feature enabled

Create a zone redundant VNET gateway in Azure Availability zones

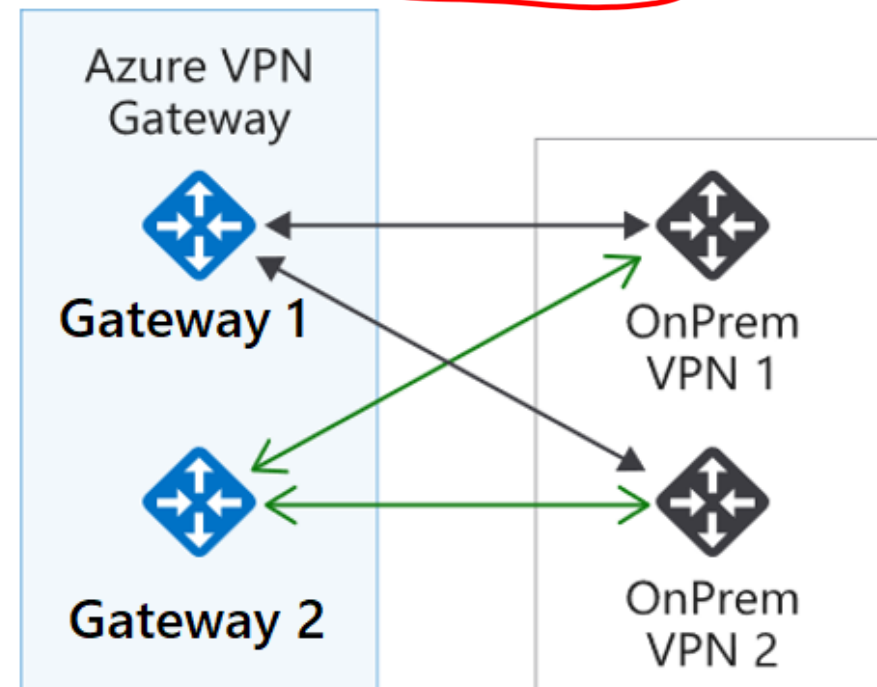


High availability options for VPN connections

Active/standby (default)



Active/active



VPN gateways are deployed
as two instances

Enable **active/active mode** for
higher availability

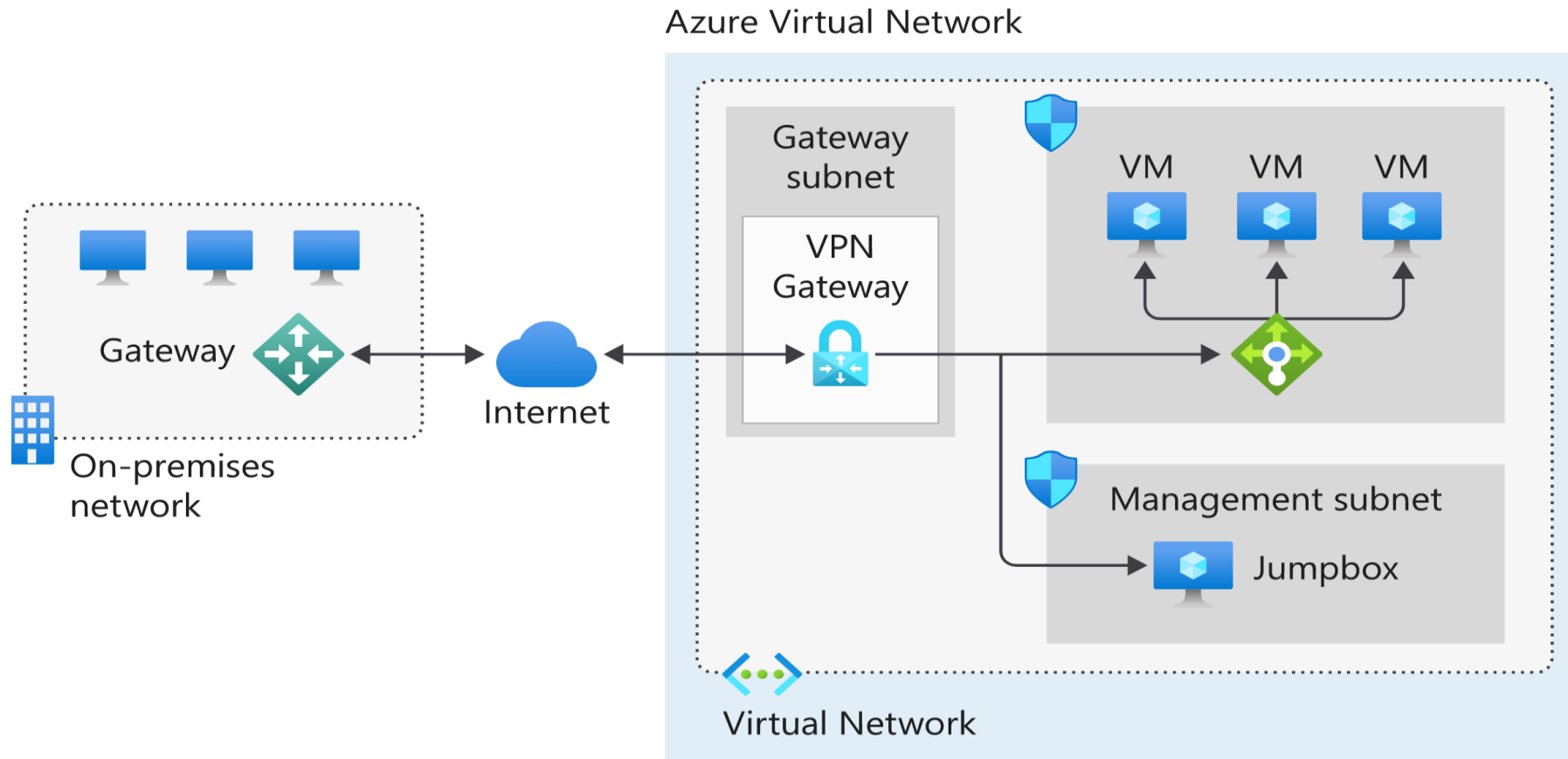
Connect Networks with Site-to-site VPN Connections



Learning Objectives – Site-to-site VPN Connections

- Site-to-site VPN Connections
- Review

Site-to-site VPN connections



Connect devices to networks with Point-to-site VPN connections



Learning Objectives – Point-to-site VPN connections

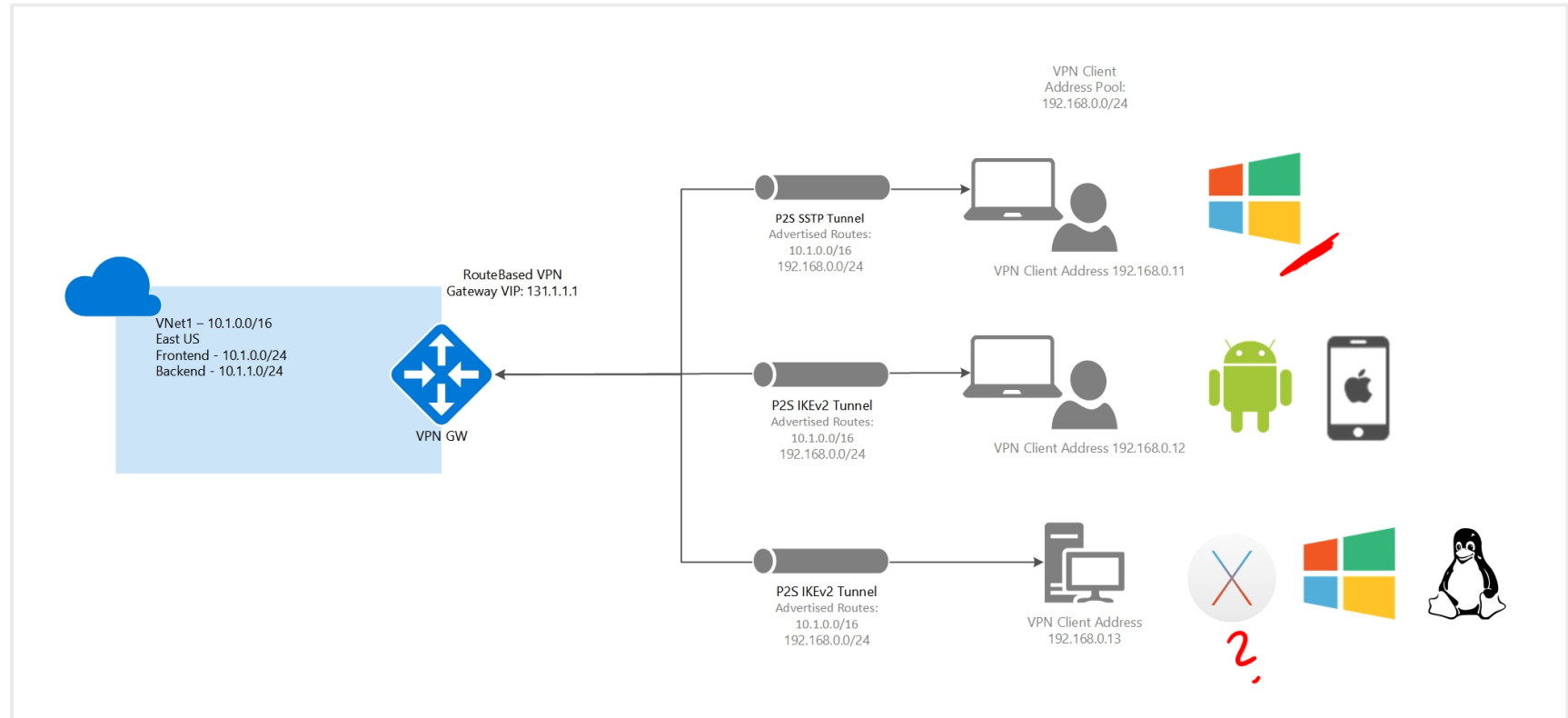
- Point-to-site protocols
- Point-to-site authentication methods
- Configure Point-to-site clients
- Learning Recap

Point-to-site protocols

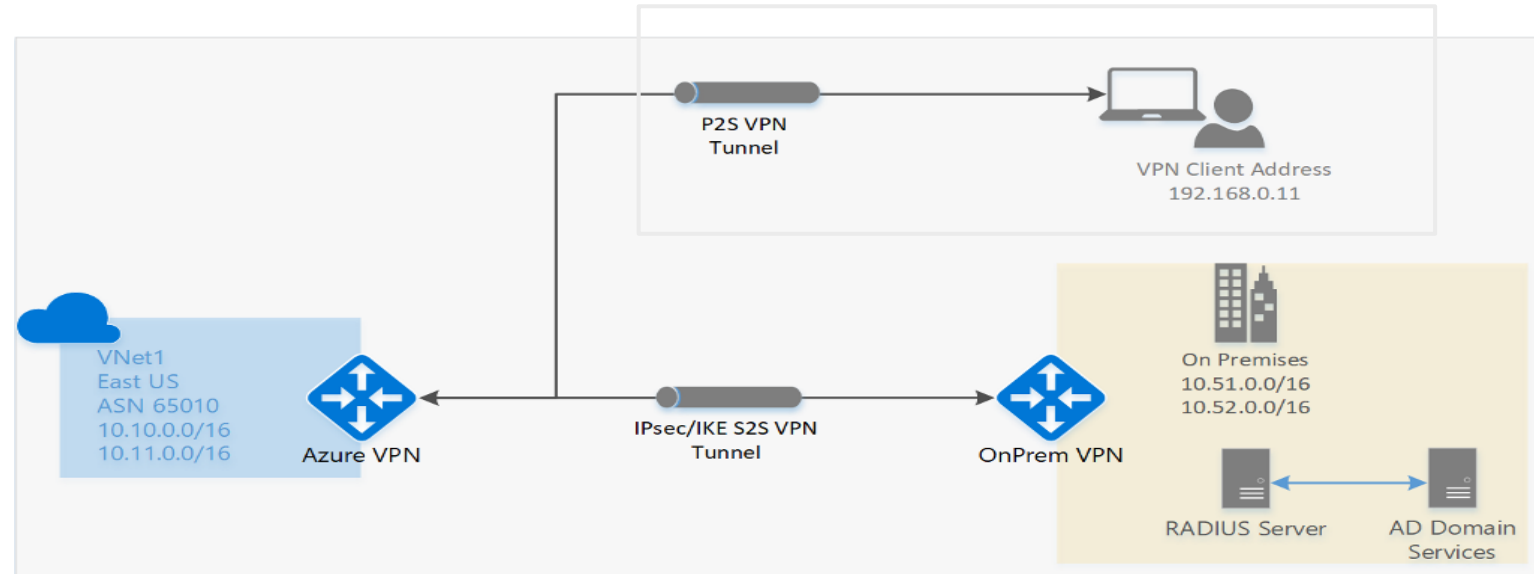
OpenVPN® Protocol

Secure Socket Tunneling Protocol (SSTP)

IKEv2 VPN



Point-to-site authentication methods



Azure certificate
authentication

Microsoft Entra
authentication

Active Directory (AD)
Domain Server

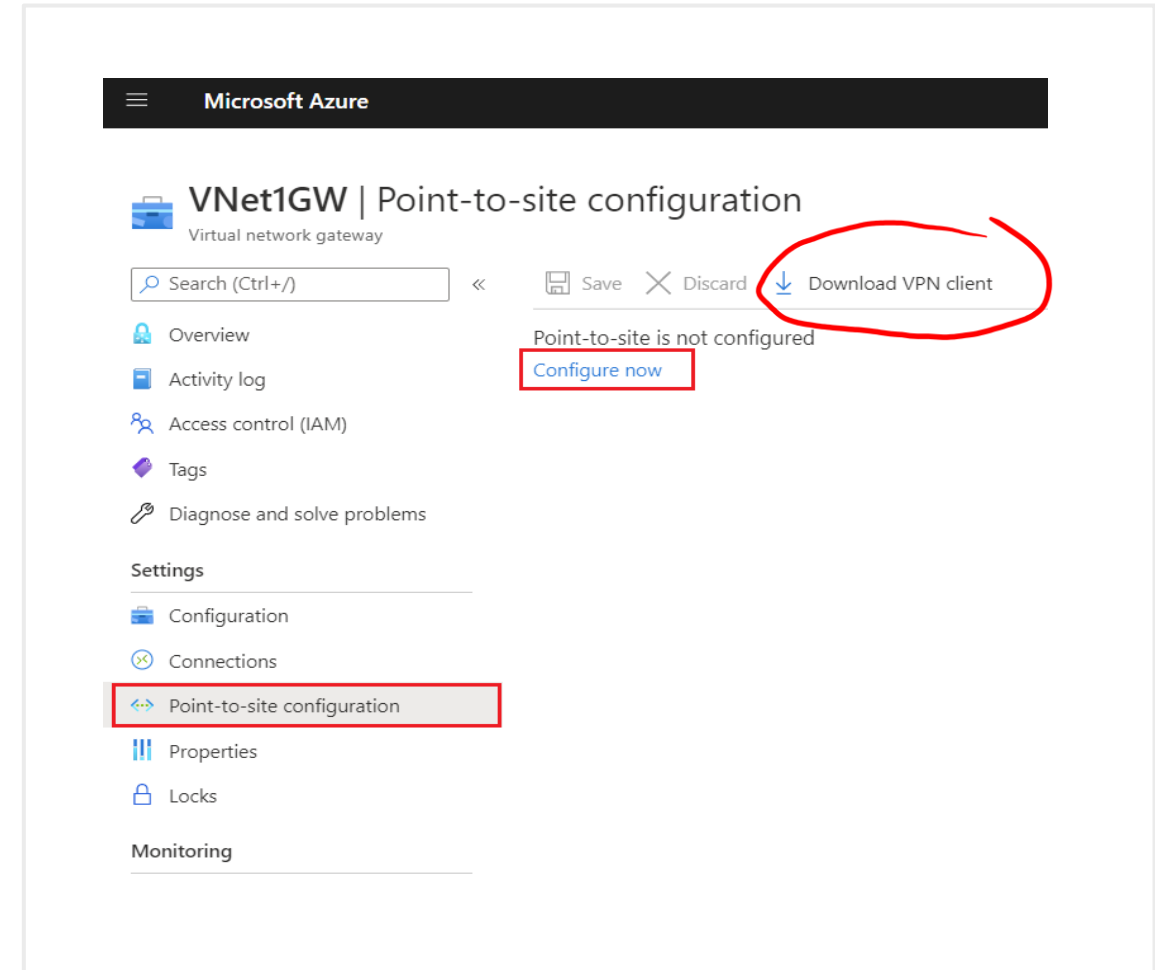
Prepare Point-to-site configuration in Azure

Navigate to the **Settings** section of the virtual network gateway page

Select **Point-to-site configuration**.
Select **Configure now** to open the configuration page

On the **Point-to-site configuration** page, in the **Address pool** box, add the private IP address range that you want to use

VPN clients dynamically receive an IP address from the range that you specify. The minimum subnet mask is 29 bit for active/passive and 28 bit for active/active configuration.





AZ-700

Tag 2

Design and Implement Hybrid Networking

Guten Morgen!

AZ-700 Agenda

github.com/www42/az-700

GH-900
😊

Module 01: Introduction to Azure Virtual Networks ✓

→ Module 02: Designing and Implementing Hybrid Networking

VWAN Fundam.

Module 03: Designing and Implementing Azure ExpressRoute

ER

Lab

Module 04: Load balance non-HTTP(S) traffic in Azure

LB

Module 05: Load balance HTTP(S) traffic in Azure

App GW

Front Door

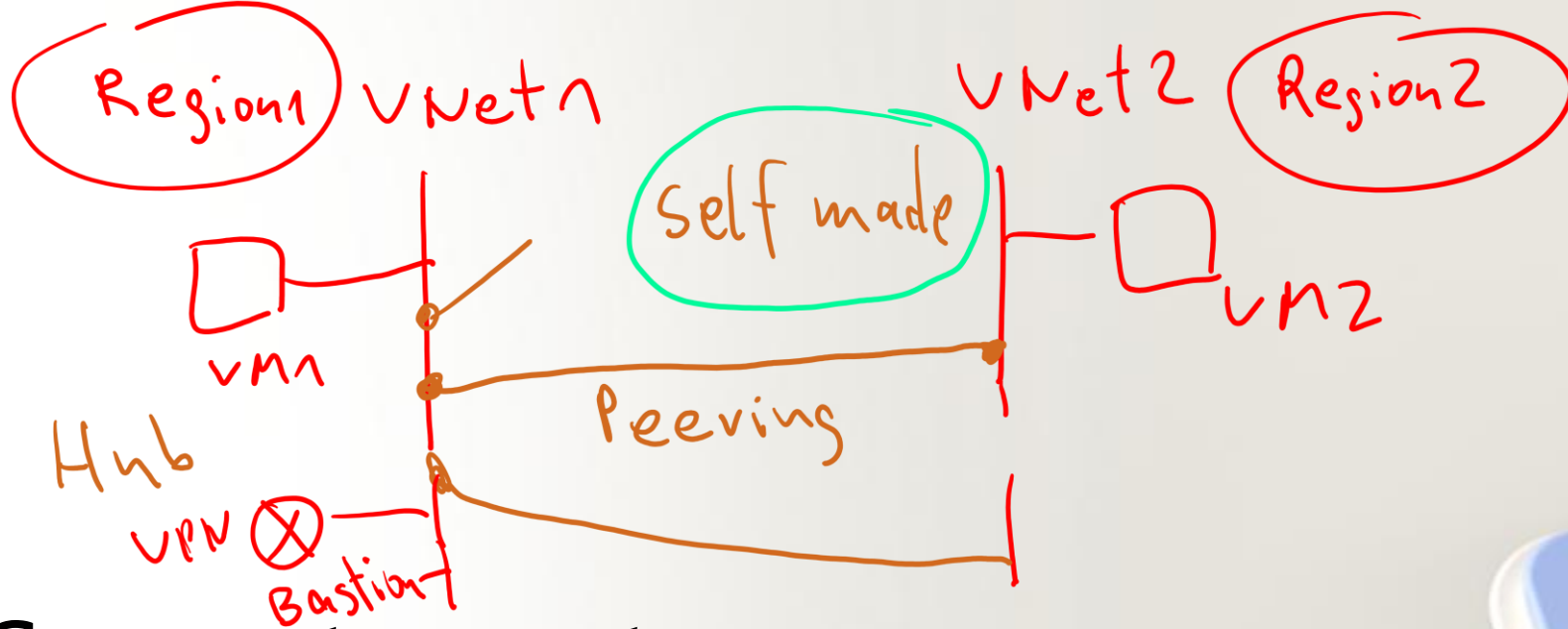
Module 06: Design and Implement Network Security

WAF

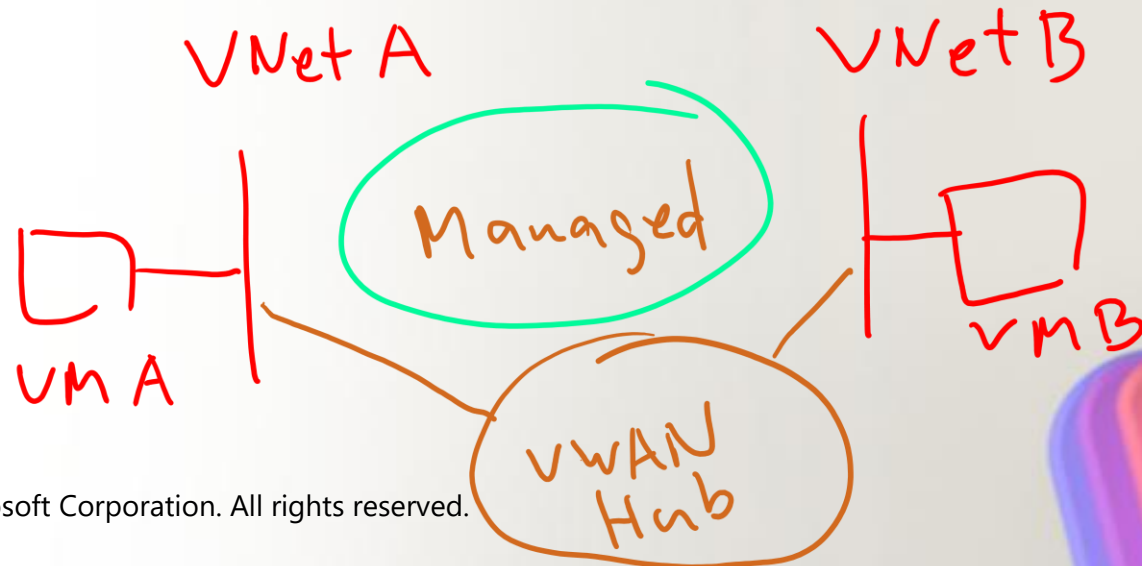
Module 07: Design and Implement private access to Azure Services

Module 08: Design and Implement Network Monitoring

12³⁰ - 13³⁰



Connect remote resources
by using Azure Virtual WANs



Learning Objectives – Azure Virtual WAN

- What is Azure Virtual WAN?
- Choose a Virtual WAN SKU
- Hub private address space
- Connect cross-tenant VNets to a virtual WAN hub
- Virtual Hub routing
- Learning Recap

What is Azure Virtual WAN?

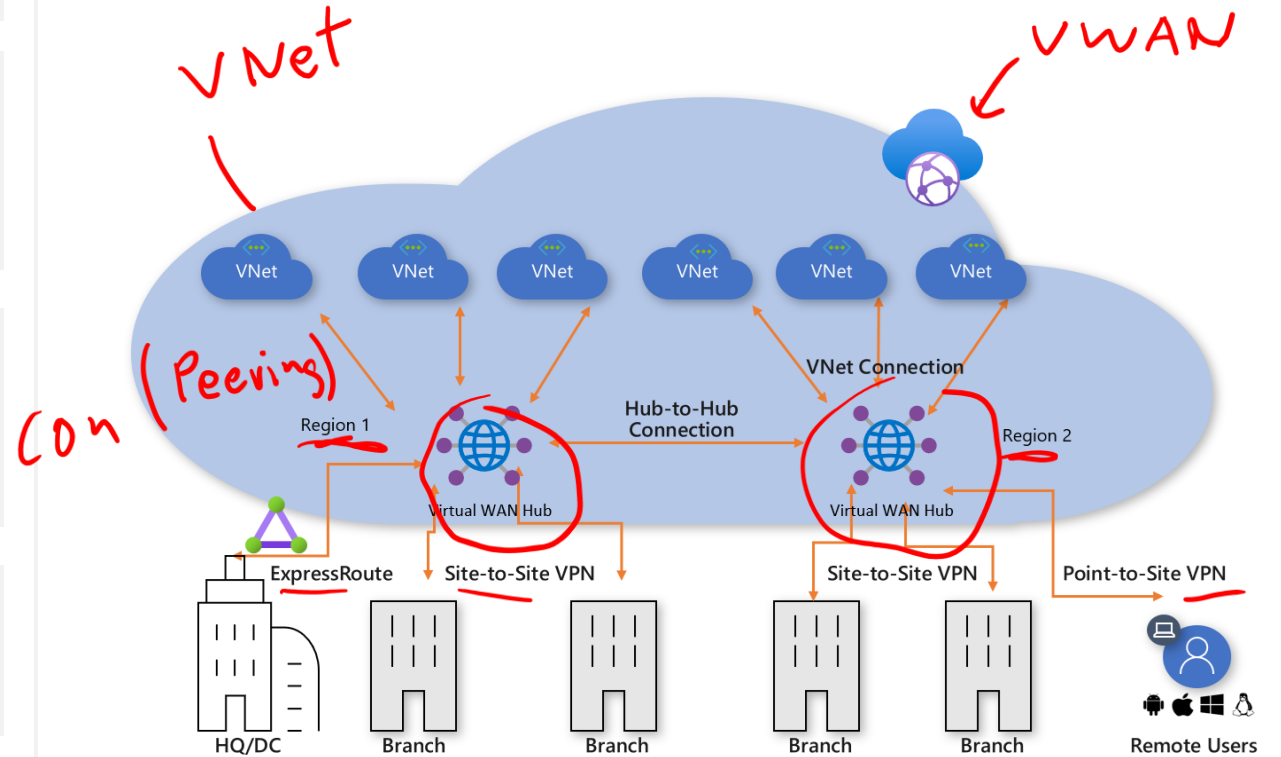
Brings together S2S, P2S, and ExpressRoute

Integrated connectivity using a hub-and-spoke connectivity model

Connect virtual networks and workloads to the Azure hub automatically

Visualize the end-to-end flow within Azure

Two types: Basic and Standard



Choose Virtual WAN SKU

Virtual WAN type	Hub type	Available configuration
Basic	Basic	Site-to-site VPN only
Standard	Standard	ExpressRoute User VPN (P2S) VPN (Site-to-site) Inter-hub and VNet-to-VNet transiting through the virtual hub

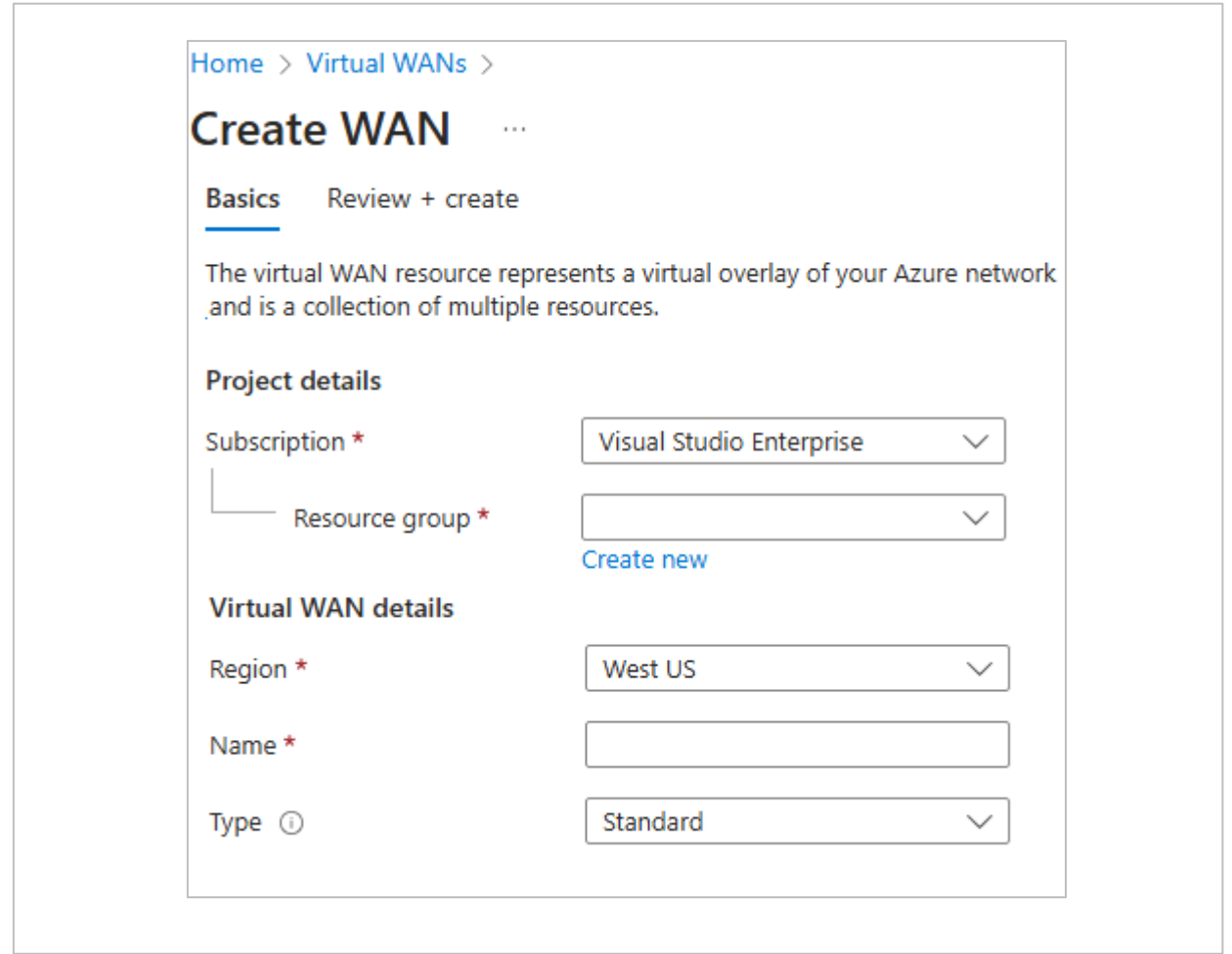
Hub private address space

Minimum address space is /24 to create a hub

No need to explicitly plan the subnet address space for the services in the virtual hub

Azure Virtual WAN is a managed service, it creates the appropriate subnets in the virtual hub for the different gateways/services

For example, VPN gateways, ExpressRoute gateways, User VPN Point-to-site gateways, Firewall, routing, etc.



The screenshot shows the 'Create WAN' page in the Azure portal. The breadcrumb navigation at the top reads 'Home > Virtual WANs >'. The main heading is 'Create WAN' followed by an ellipsis. Below the heading are two tabs: 'Basics' (which is selected and underlined) and 'Review + create'. A descriptive text states: 'The virtual WAN resource represents a virtual overlay of your Azure network and is a collection of multiple resources.'

The 'Project details' section contains two dropdown menus: 'Subscription *' with 'Visual Studio Enterprise' selected, and 'Resource group *' which is currently empty. Below the 'Resource group' dropdown is a blue link that says 'Create new'.

The 'Virtual WAN details' section contains three dropdown menus: 'Region *' with 'West US' selected, 'Name *' which is empty, and 'Type ⓘ' with 'Standard' selected.

(VWAN) ①

Virtual Hub Routing

Hub route
table

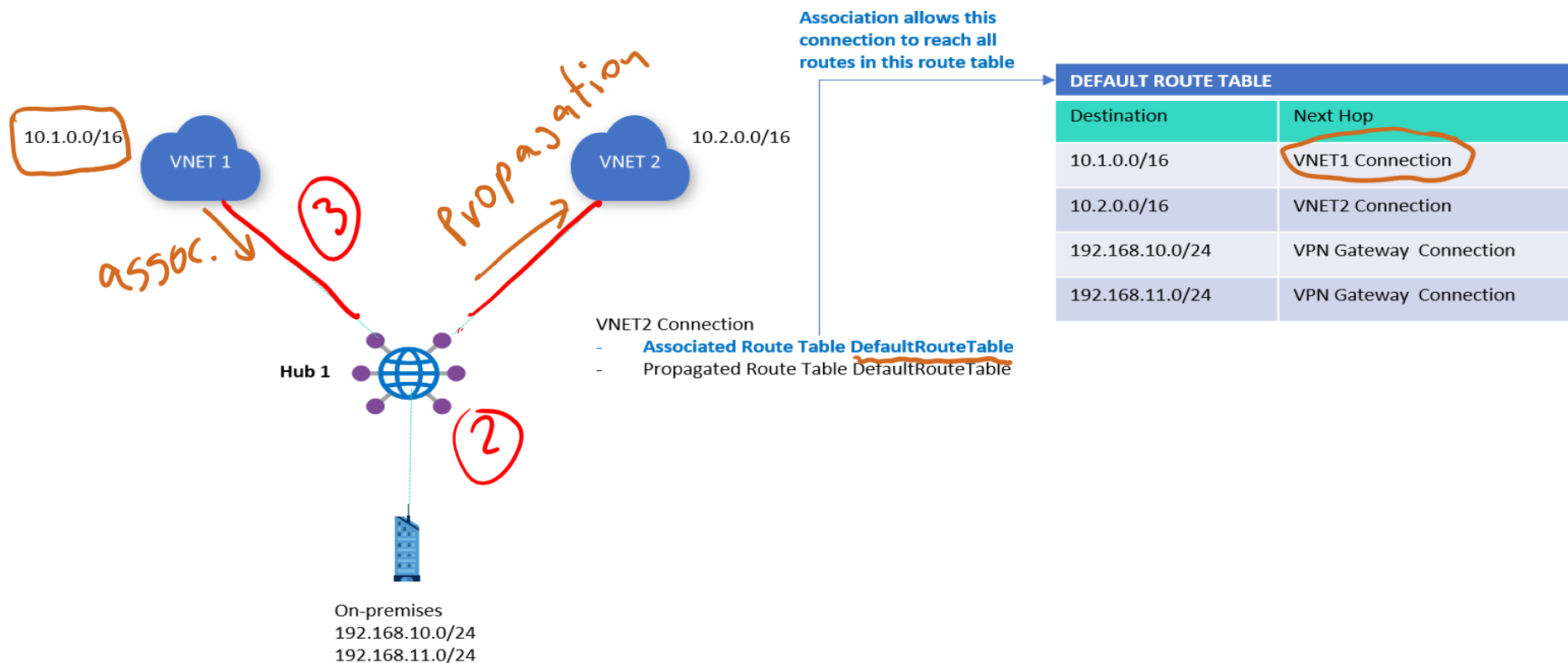
Connections

Association

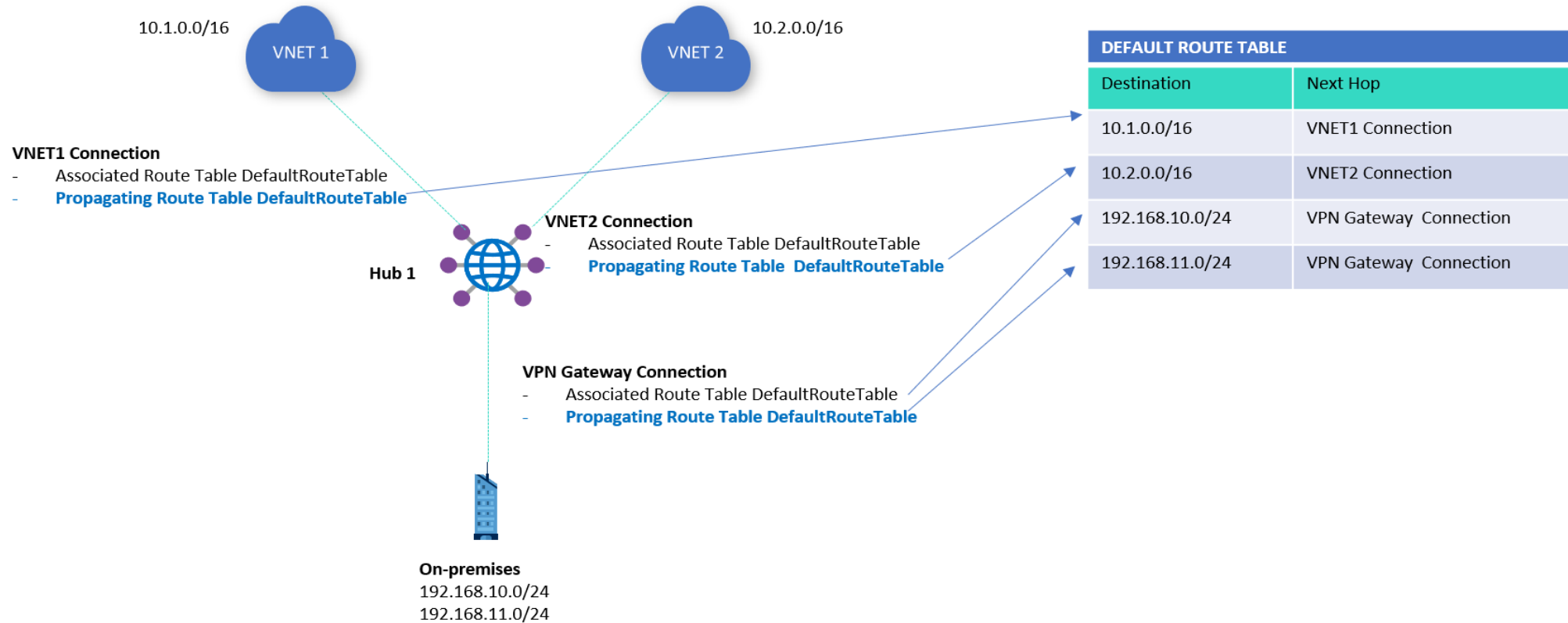
Propagation

Labels

Static routes



Virtual Hub Routing – continued



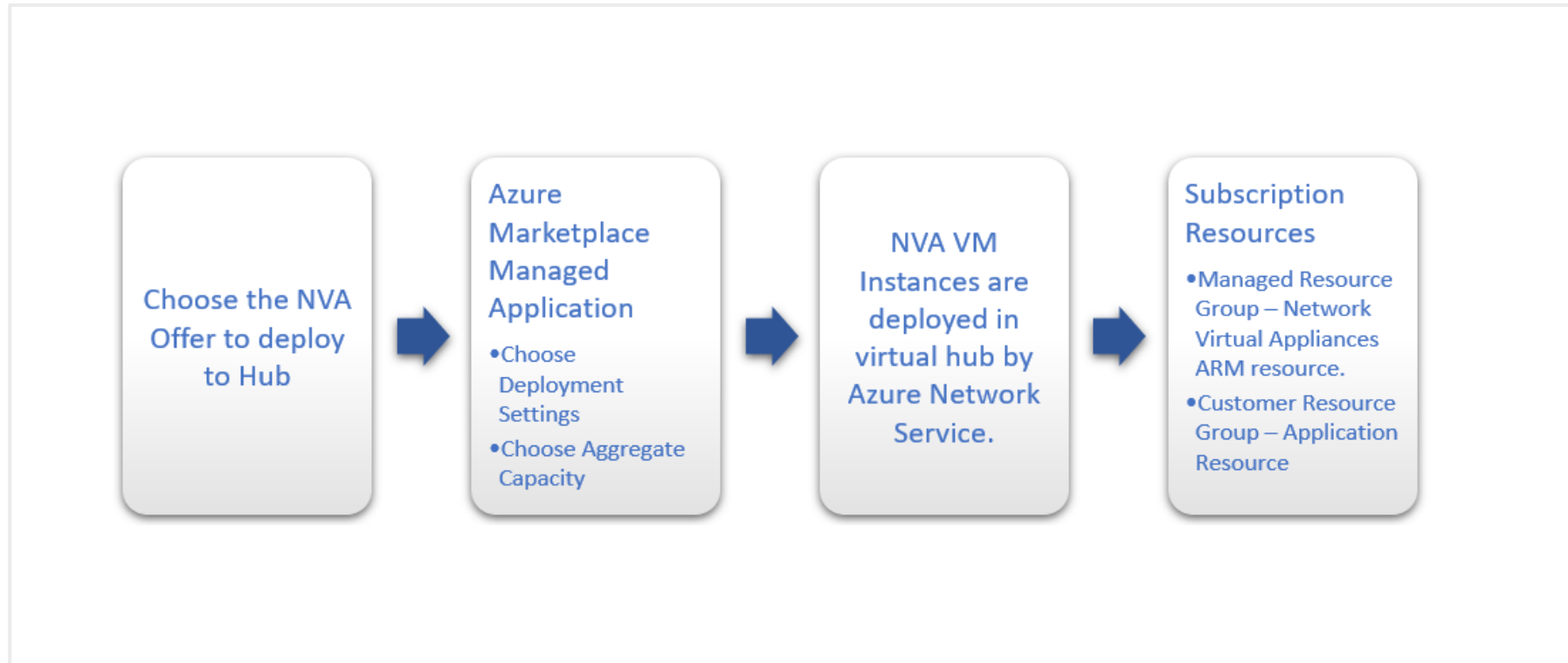
Create a network virtual appliance (NVA) in a virtual hub



Learning Objectives – NVA in a Virtual Hub

- Manage an NVA in a Virtual Hub
- Deploy an NVA in your Virtual Hub
- Learning Recap

Manage an NVA in a Virtual Hub



Deploy an NVA in your Virtual Hub

Locate the Virtual WAN hub you created in the previous step and open it

Find the Network Virtual Appliances tile and select the Create link.

On the **Network Virtual Appliance** blade, select your preferred provider based on available selections, then select the **Create** button

Network Virtual Appliance

arubaedgeconnectenterprise

barracudasdwanrelease

checkpoint 

ciscosdwan 

fortinet-ngfw 

fortinet-sdwan-and-ngfw

fortinet-sdwan

fortinet

versanetworks

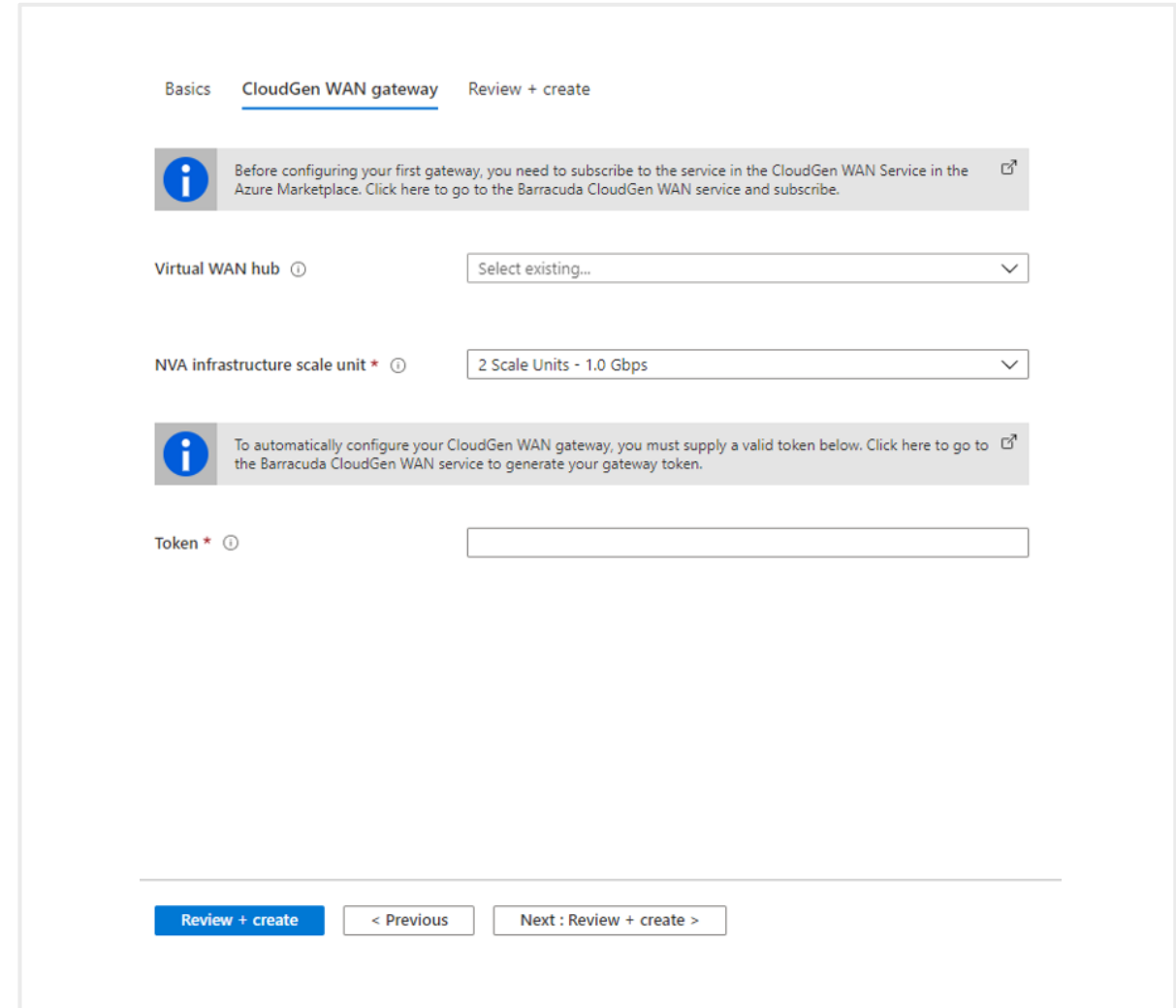
vmwaresdwaninvwan

Deploy an NVA in your Virtual Hub Cont.

Virtual WAN Hub - The Virtual WAN hub you want to deploy this NVA into

NVA Infrastructure Units - Indicate the number of NVA Infrastructure Units you want to deploy this NVA with. Choose the amount of aggregate bandwidth capacity you want to provide across all the branch sites that will be connecting to this hub through this NVA.

Token - Barracuda requires that you provide an authentication token here in order to identify yourself as a registered user of this product. You'll need to obtain this from Barracuda.



The screenshot shows the 'CloudGen WAN gateway' configuration page in the Azure portal. The page has three tabs: 'Basics', 'CloudGen WAN gateway' (which is selected), and 'Review + create'. Below the tabs, there are two informational messages: one about subscribing to the service in the Azure Marketplace, and another about providing a valid token for automatic configuration. The configuration fields include 'Virtual WAN hub' (a dropdown menu with 'Select existing...' as the current selection), 'NVA infrastructure scale unit' (a dropdown menu with '2 Scale Units - 1.0 Gbps' as the current selection), and 'Token' (a text input field). At the bottom of the page, there are three buttons: 'Review + create' (highlighted in blue), '< Previous', and 'Next : Review + create >'.

Lab 2:

Create and configure a
Virtual Network Gateway

Jetzt

Create a virtual WAN

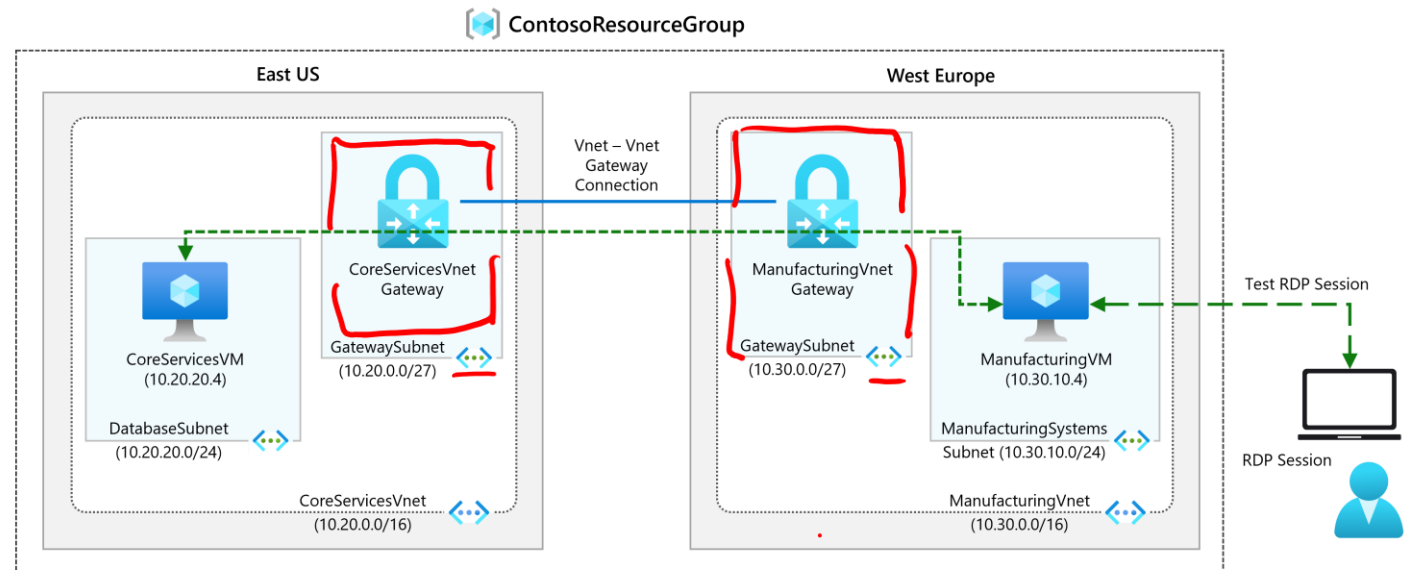
Morgen



Exercise – Create and Configure a Virtual Network Gateway



Configure a virtual network gateway to connect the Contoso Core Services VNet and Manufacturing VNet



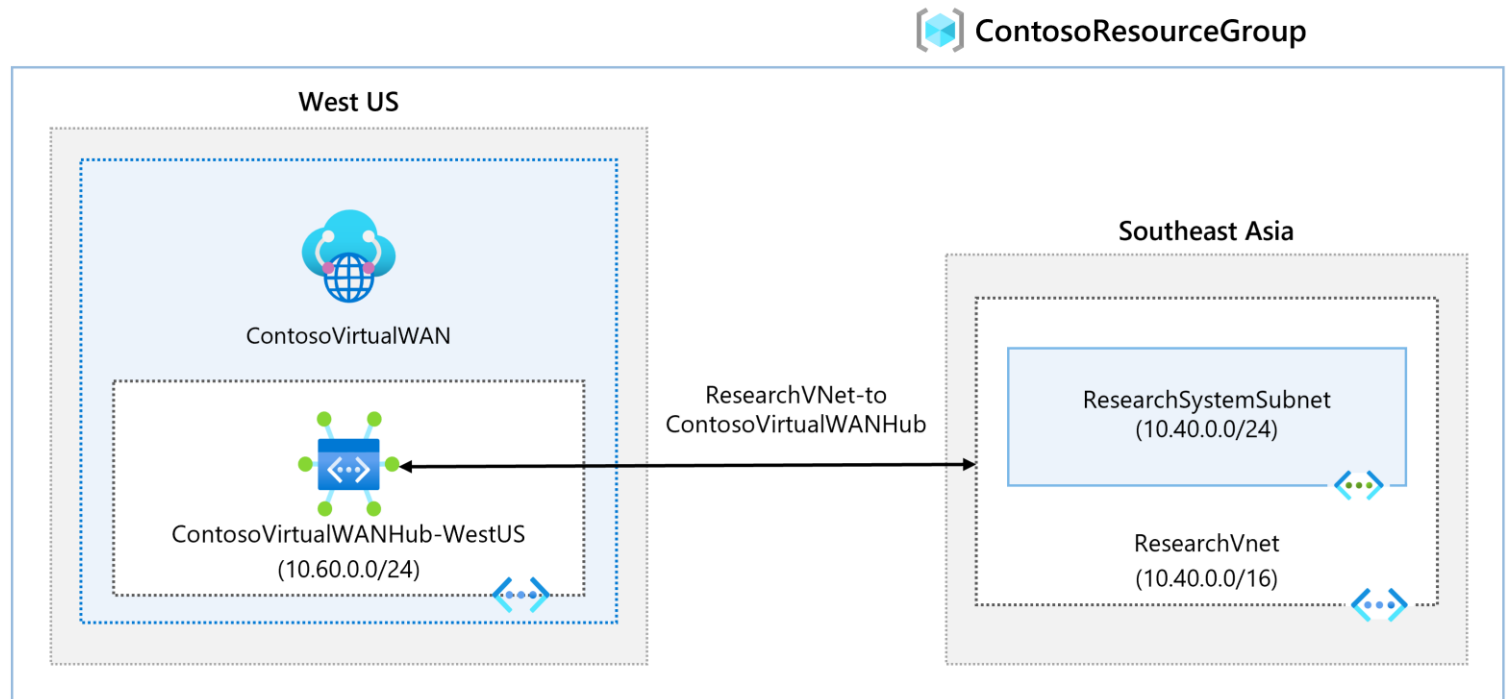
Exercise – Create a Virtual WAN by Using Azure Portal



Task 1: Create a Virtual WAN

Task 2: Create a hub

Task 3: Connect a VNet to the Virtual Hub



End of presentation

